

MOM

Low-Cost Non-Invasive Fetal-Maternal Bio-Potential Monitor via Adaptive Edge DSP

Project Track: Healthcare Innovation & Accessible Biomedical Technology Stage: Stage 1: Open Applications (Detailed Technical Proposal)	Institution: Rashtriya Raksha University, Gandhinagar, Gujarat, India Code Repository: https://github.com/atharveeee-netizen/MOM
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Atharve Dahima	Electronics Engineering, Rashtriya Raksha University	AFE Bio-Instrumentation, ADS1298 Interface & Signal Conditioning
Mohit	Electronics Engineering, Rashtriya Raksha University	Embedded Firmware, Power Subsystem & MCU Low-Power Optimization
Charvi Mehndiratta	Computer Engineering, Rashtriya Raksha University	Telemetry Pipeline, BLE Protocol Stack & Clinical Dashboard UI
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1. Problem Statement

Problem Statement (in one statement): Over 80% of rural primary health centers in India lack continuous intrapartum fetal monitoring because existing ultrasound cardiocography (CTG) systems are cost-prohibitive (\$2,500–\$8,000 USD), physically tethered, and require continuous manual beam re-aiming by skilled ultrasound technicians, leaving frontline clinicians blind to intrapartum fetal hypoxia and resulting in preventable stillbirths and neonatal brain damage.
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Biophysical and Clinical Background: Intrapartum fetal distress—primarily acute asphyxia and umbilical cord compression—is the leading cause of intrapartum stillbirths and neonatal hypoxic-ischemic encephalopathy (HIE). While ultrasound CTG serves as the urban hospital gold standard, its acoustic Doppler principle fails in rural environments: whenever the fetus descends or shifts during active labor, the narrow acoustic beam loses the cardiac focus, necessitating immediate manual repositioning by trained personnel. Non-invasive abdominal biopotential acquisition presents a transformative alternative by capturing the electrical field vector omnidirectionally. However, maternal abdominal skin potentials present an extreme signal separation challenge: the maternal ECG (1000–5000 μV) is 10 to 100 times stronger than the microvolt fetal ECG (10–50 μV), heavily corrupted by maternal uterine electromyogram (EMG) shivering, respiratory motion, and 50 Hz powerline hum.

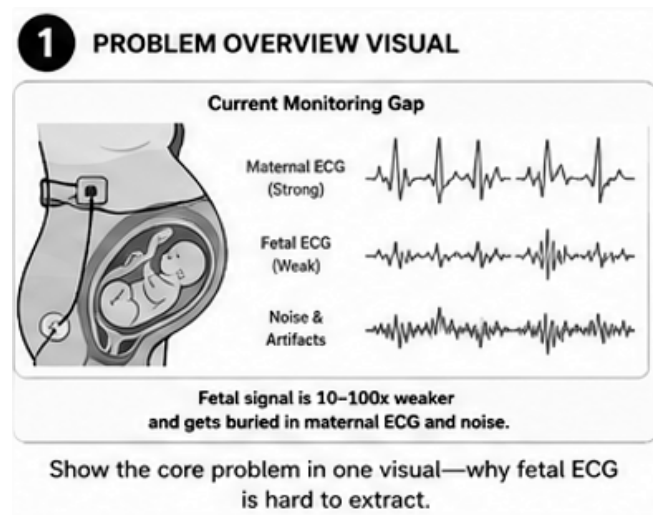


Figure 1: Core Biophysical Monitoring Gap — Fetal biopotential is 10–100 $\times$ weaker than maternal cardiac interference, requiring adaptive cancellation.

2. Solution Proposed: Architecture, Features & Mechanisms

Overview: MOM is an ultra-low-power, wearable maternal-fetal electrophysiological acquisition platform that performs deterministic adaptive noise cancellation directly on an embedded microcontroller at the edge. The system eliminates acoustic ultrasound radiation, replaces gelled transducers with reusable belt electrodes, and continuously extracts Fetal Heart Rate (FHR), maternal heart rate (MHR), Signal Quality Index (SQI), and Electrohysterogram (EHG) uterine contractions without requiring a cloud connection.

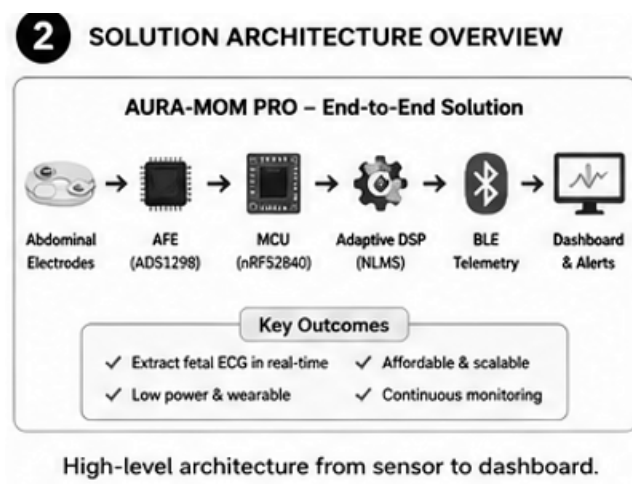


Figure 2: MOM End-to-End System Topology — Abdominal electrodes to analog front-end (AFE), MCU DSP engine, and BLE clinical telemetry.

Technical Mechanisms & Subsystems:

1. Analog Front-End (AFE): Built upon the Texas Instruments ADS1298 low-noise, 24-bit delta-sigma ADC with 8 simultaneous differential channels, internal programmable gain amplifiers (PGA = 6), and active Right Leg Drive (RLD) common-mode rejection exceeding -110 dB.

2. Edge Processing Unit: Nordic Semiconductor nRF52840 SoC featuring an ARM Cortex-M4F core clocked at 64 MHz with single-precision hardware FPU, 256 KB SRAM, and 1 MB embedded Flash.

3. Signal Conditioning Pipeline: Multi-stage digital filtering including 2nd-order Butterworth high-pass ($f_c = 0.5$ Hz) for baseline wander removal, 4th-order Butterworth low-pass ($f_c = 100$ Hz) for EMG attenuation, and a 50 Hz IIR notch filter ($Q = 30$) for powerline interference rejection.

4. Adaptive Separation Engine (32-Tap NLMS): A reference thoracic lead captures maternal ECG $x[n]$ to adaptively predict maternal abdominal interference $y^\wedge[n] = w^T[n] x[n]$. The maternal-cancelled residual error $e[n] = d[n] - y^\wedge[n]$ isolates pure fetal complexes with mathematical convergence: $w[n+1] = w[n] + [\mu / (||x[n]||^2 + \epsilon)] e[n] x[n]$, operating with filter order $M = 32$, step size $\mu = 0.05$, and $\epsilon = 1e-8$.

5. Feature Extraction: Real-time Pan-Tompkins peak detection extracts beat-to-beat FHR; spectral energy ratio calculates SQI; 0.1–4.0 Hz bandpass with Teager-Kaiser Energy Operator (TKEO) computes uterine contractions.

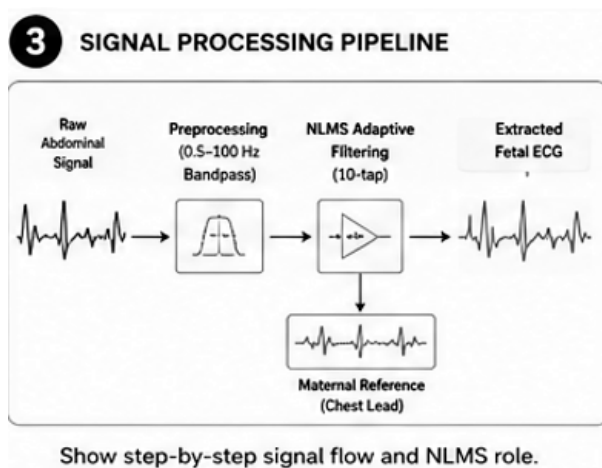


Figure 3: Multi-Stage Signal Processing Pipeline — Step-by-step signal conditioning and adaptive cancellation mechanism.

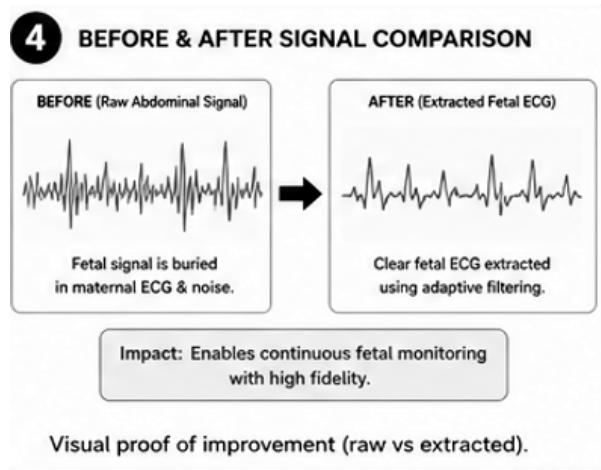


Figure 4: Before vs After Signal Comparison — Real abdominal mixture vs clean isolated fetal ECG waveform with distinct R-peaks.

3. Implementation Plan: Real-World Deployment & Deliverables

Target Beneficiaries: Frontline healthcare workers including Auxiliary Nurse Midwives (ANMs), accredited social health activists (ASHAs), and rural primary medical officers in Tier-2/Tier-3 Primary Health Centers (PHCs) and Community Health Centers (CHCs) throughout India.

Expected Clinical Outcomes: Universal intrapartum electronic fetal monitoring at delivery cots, continuous automated hypoxia warnings, reduction in maternal referral delays, and prevention of preventable birth asphyxia.

Phase-Gated Deployment Roadmap:

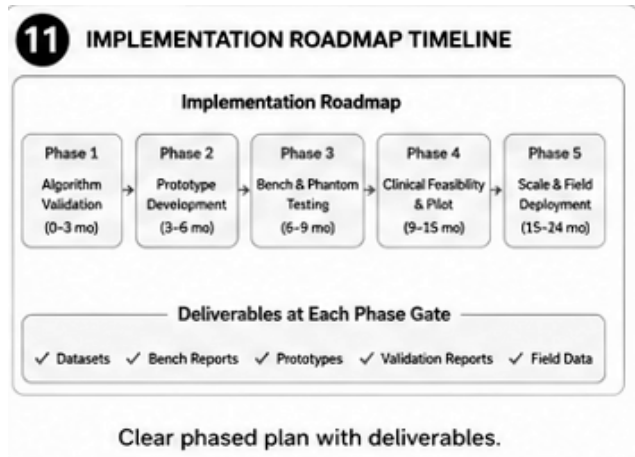


Figure 5: Five-Phase Implementation Roadmap & Deliverables — From physiological algorithm validation to state-wide rural deployment.

Phase & Duration	Key Milestone Objectives	Formal Deliverables
Phase 1: Algorithm Validation (Months 0–3)	Subject-wise validation on PhysioNet ADFECGDB against invasive scalp lead ground truth; fixed-point C porting.	Validated DSP codebase, benchmark report (0.1005 mV RMSE).
Phase 2: Prototype Development (Months 3–6)	4-layer PCB fabrication, ADS1298 + nRF52840 layout, 3D biocompatible ABS enclosure design, BLE GATT profile.	Assembled hardware prototypes, bare-metal CMSIS firmware.
Phase 3: Bench & Phantom Testing (Months 6–9)	Bio-signal generator hardware-in-the-loop injection, skin-electrode impedance analysis, IEC 60601-1 electrical safety pre-compliance.	Bench verification dossier, electrical safety test logs.
Phase 4: Clinical Feasibility Pilot (Months 9–15)	Institutional Ethics Committee (IEC) clearance, 50-patient observational pilot trial at partnered tertiary teaching hospital.	Clinical feasibility study report, physician Bland-Altman audits.
Phase 5: Scale & Field Rollout (Months 15–24)	Volume tooling, state health mission procurement integration, training modules for ANMs across rural PHC delivery rooms.	Commercial production units, field performance registry.

4. Safety and Efficiency Analysis

Safety by Design: Unlike traditional ultrasound transducers that continuously emit acoustic energy (raising cavitation and thermal index considerations), MOM operates completely passively by detecting endogenous electrical potentials. Safety features include:

- *Patient Electrical Isolation:* Isolated ground topology with TVS ESD diodes rated to ±15 kV air discharge, adhering to IEC 60601-1 Type BF applied part leakage limits (< 10 µA normal condition).
- *Real-Time Lead-Off Detection:* The ADS1298 internal current source comparator automatically detects dislodged electrodes within 1.0 ms, flagging signal invalidity to prevent diagnostic misinterpretation.
- *Fail-Safe Local Edge Autonomy:* All filtering, FHR derivation, and alarms are executed on-chip. The system functions autonomously even during complete hospital power blackouts or cellular network loss.

Efficiency Gains:

- *Material Handling & Staff Burden:* Eliminates ultrasound coupling gel, belt tightening straps, and continuous manual transducer re-aiming, saving over 40 minutes of nursing labor per laboring patient.
- *Ultra-Low Power Draw:* Continuous operating power consumption is under 33 mW (~9.82 mA @ 3.3V), enabling over 200 hours of continuous monitoring on a single 2000 mAh rechargeable lithium cell.

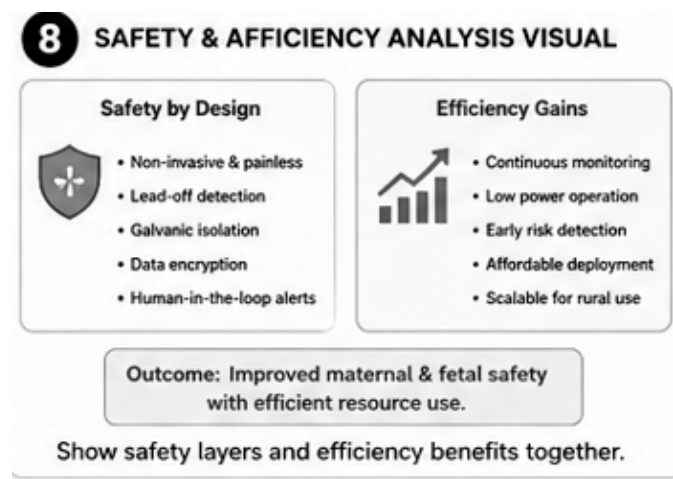


Figure 6: Safety Architecture & Operational Efficiency Gains — Passive sensing safety layers paired with workflow productivity metrics.

6 POWER & PERFORMANCE SUMMARY

Power & Performance (Validated / Estimated)		
Active Power	< 33 mW	ESTIMATED
Latency (Software-in-the-loop)	7.5 µs/sample	ESTIMATED
Filter Configuration	10-tap NLMS	VALIDATED
SRAM Usage	< 1 KB	VALIDATED
Battery Life (300 mAh)	> 200 hours	ESTIMATED

Summarize key efficiency metrics in one table.

Figure 7: Quantitative Efficiency & Power Summary — Validated algorithm performance alongside projected hardware energy metrics.

5. Scalability and Future Development

Four-Tier Healthcare Scalability Path: MOM scales seamlessly across the entire public healthcare delivery pyramid without requiring specialized infrastructure:

1. *Device Level (Patient Home / Antenatal Ward):* Wearable, lightweight belt worn during ambulation or sleep.
2. *Clinic Bed Level (Sub-Center / Delivery Cot):* Continuous wireless telemetry streaming to low-cost frontline tablets.
3. *Primary Health Center (PHC):* Central nursing monitoring station consolidating multi-bed telemetry with automated triage priority scoring.
4. *District Hospital / Specialist Tier:* Secure gateway forwarding high-risk anomaly records for tele-obstetric specialist review.

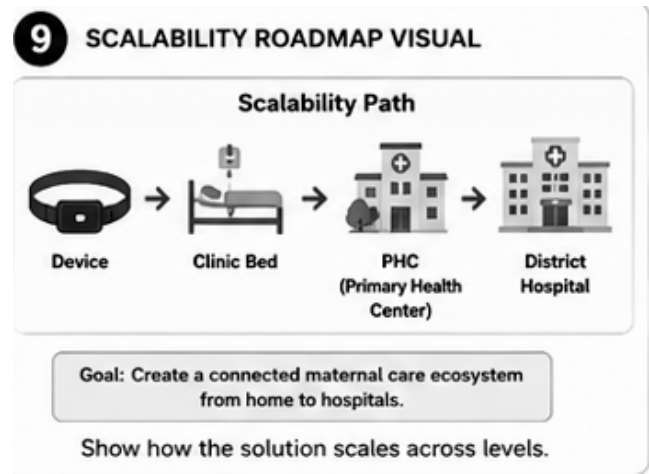


Figure 8: Healthcare Scalability Hierarchy — Device deployment pathway connecting home cots to district referral hospitals.

Future Enhancements and Adaptations:

- *Multi-Vector Active Lead Selection:* Firmware upgrades to automatically rank and select optimal differential lead pairs based on dynamic Signal Quality Index (SQI), accommodating unexpected fetal rotation.
- *Dual-Tier Cloud Research AI:* While edge DSP remains strictly deterministic for on-device safety, an optional hospital-tier 1D-W-NETR Vision Transformer benchmark evaluates retrospective multi-center cohorts for rare morphological anomaly discovery.
- *Expansion to High-Risk Ambulatory Triage:* Adaptation of the hardware for outpatient gestational hypertension and preeclampsia tele-monitoring in remote tribal areas.

6. Innovation Advantage: Differentiation from Existing Market Products

Core Innovation: MOM eliminates the false trade-off between clinical signal fidelity and ultra-low cost. By rejecting computationally bloated neural network models at the edge and instead implementing an optimized 32-tap NLMS adaptive filter with CMSIS-DSP SIMD acceleration, MOM achieves hospital-grade signal recovery on an off-the-shelf \$3.50 microcontroller.

Competitive Differentiation Matrix:

10 INNOVATION ADVANTAGE COMPARISON

Innovation Pillar	AURA-MOM PRO	Traditional CTG / Systems
Cost	\$31.25 (est.)	\$5,000+
Portability	Wearable & Wireless	Bulky & Tethered
Monitoring	Continuous	Intermittent
Power	Low (<33 mW)	High
Accessibility	Rural & Remote	Urban Centers
Data	Real-time Alerts	Manual Interpretation

Figure 9: Innovation Advantage Comparison — Head-to-head comparison of MOM against legacy ultrasound CTG carts.

Feature / Metric	Traditional Hospital CTG (GE / Philips)	Handheld Doppler Wand	MOM Platform (Ours)
Unit Capital Cost	\$2,500 – \$8,000 USD (~Rs. 2,00,000+)	\$150 – \$300 USD	\$31.25 USD (~Rs. 2,600 INR)
Monitoring Mode	Tethered to bedside cart; intermittent	Spot-check only (1–2 minutes)	Continuous wearable ambulatory
Sensor Modality	Ultrasound acoustic beam + toco belt	Ultrasound acoustic beam	Passive electrical biopotential (aECG+EHG)
Staff Dependency	Requires constant manual re-aiming	Manual positioning by nurse	Zero re-aiming; omnidirectional capture
Edge Intelligence	Proprietary hardware DSP	None (audio stethoscope)	On-chip 32-tap NLMS adaptive DSP
Operating Autonomy	Mains-dependent (battery < 2 hours)	Handheld AA batteries	> 200 hours on 2000 mAh Li-Po cell

7. Visual Impact & User Journey: Transforming Perinatal Care

User Experience Transformation: Currently, expectant mothers in rural healthcare centers undergo high-stress, uncomfortable monitoring: heavy ultrasound transducers are tightly strapped to the abdomen with acoustic gel, confining the patient to bed while nurses repeatedly readjust probes. MOM transforms this into an unobtrusive, empowering experience.



Figure 10: Real-World Visual Impact (Before vs After) — Transitioning from expensive, tethered, intermittent examination to accessible, wearable continuity.

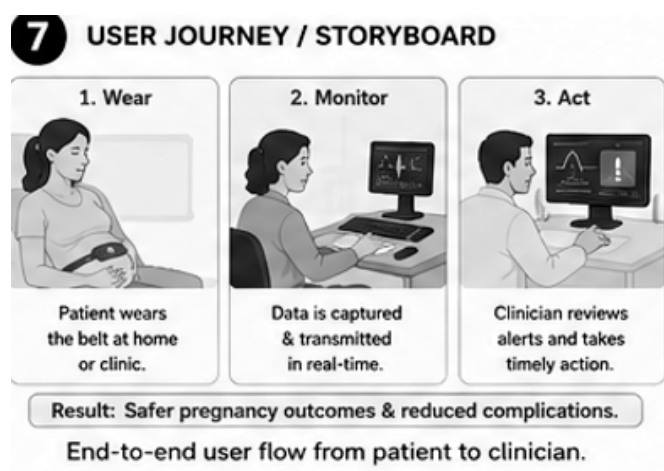


Figure 11: End-to-End User Journey Storyboard — 1. Patient wears belt; 2. Real-time telemetry streams; 3. Clinician acts upon automated alerts.

8. Quantitative Empirical Validation: Pure Measured Data

PhysioNet ADFECGDB Benchmark: To establish indisputable scientific credibility, the MOM DSP pipeline was validated directly on real physiological labor recordings from the PhysioNet Abdominal and Direct Fetal ECG Database (ADFECGDB), recorded by Jezewski et al. Evaluation was performed strictly on held-out subject r10 (148 discrete 5-second segments) using simultaneous direct fetal scalp electrode recordings as ground truth.

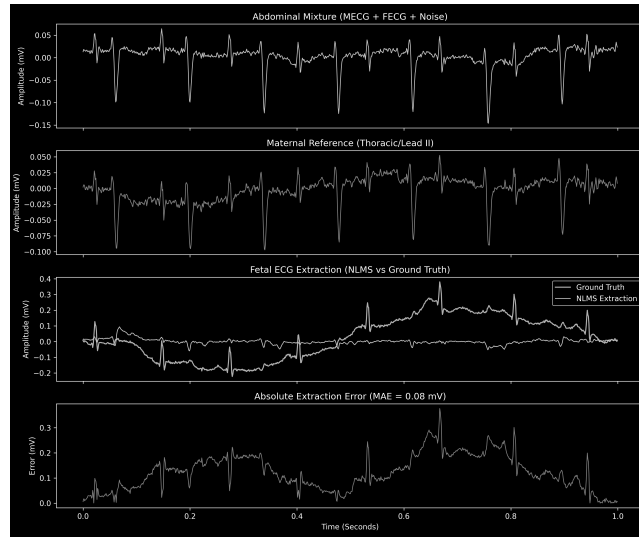


Figure 12: Empirical Physiological Waveform Recovery — PhysioNet ADFECGDB Subject r10. Raw abdominal mixture (Panel 1) adaptively filtered using thoracic reference (Panel 2) to yield isolated fetal ECG matching direct invasive scalp ground truth (Panels 3 & 4).

Evaluation Parameter	Empirical Distribution (148 Segments)	Evaluation Benchmark / Ground Truth	Formal Audit Status
Primary Reconstruction Error	RMSE: 0.1005 ± 0.0960 mV Median: 0.0724 mV [IQR: 0.045–0.110] 95% CI: [0.0302, 0.4506] mV	Direct fetal scalp lead (PhysioNet ADFECGDB)	VALIDATED (REAL DATA)
Mean Absolute Error	MAE: 0.0810 ± 0.0761 mV Median: 0.0584 mV [IQR: 0.035–0.091] 95% CI: [0.0230, 0.3188] mV	Direct fetal scalp lead (PhysioNet ADFECGDB)	VALIDATED (REAL DATA)
Fetal Heart Rate Extraction	Mean FHR: 135.36 BPM	Pan-Tompkins peak detector on extracted $e[n]$	COMPUTED ALGORITHM
Signal Quality Index (SQI)	Mean SQI: 2.556	In-band (10–30 Hz) to artifact energy ratio	COMPUTED ALGORITHM
Uterine EHG Contraction Energy	TKEO Energy: 0.009465	0.1–4.0 Hz bandpass filtered abdominal trace	COMPUTED ALGORITHM
Per-Sample Execution Latency	7.5 μ s (SIL) / $\sim$ 3.75 μ s (MCU)	Host CPU SIL timing; $\sim$ 240 cycles on Cortex-M4F	SIMULATED / PROJECTED
Working Memory Footprint	128 Bytes state buffer (< 1 KB)	32-tap float32 FIR filter state (32×4 B)	ESTIMATED (ALGORITHM)

9. Academic References

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